

PYTHON FUNDAMENTALS – FINAL NOTES & TASKS

1. VARIABLES

Variables are used to store data.

Python does not require explicit data type declaration.

Example:

```
name = "Ravi"
```

```
age = 18
```

```
marks = 95.5
```

Rules:

- Must start with a letter or underscore (_)
- Cannot start with a number
- Case-sensitive (age ≠ Age)

2. DATA TYPES

Type Example

```
int 10
```

```
float 5.5
```

```
str "Hello"
```

```
bool True
```

Example:

```
x = 10
```

```
y = 5.5
```

```
name = "Python"
```

```
is_passed = True
```

Check type:

```
print(type(x))
```

3. COMMENTS

Single-line:

This is a comment

Multi-line:

```
"""
```

This is a
multi-line comment

```
"""
```

4. INPUT & OUTPUT

```
name = input("Enter your name: ")  
print(name)
```

5. STRINGS

Strings are text data.

Example:

```
text = "Python"
```

6. STRING INDEXING

<pre>P y t h o n 0 1 2 3 4 5</pre>
--

Access:

text[0] → P

text[1] → y

text[-1] → n

PRACTICE TASKS

Task 1: Variable Program

Create variables: Name, Age, Subject and print them.

Task 2: Data Types

Create int, float, string, bool and print their types.

Task 3: Input Program

Take name and age and print:

Hello

You are years old

Task 4: Comments

Use 2 single-line and 1 multi-line comment.

Task 5: Student Info

Print:

--- Student Info ---

Name: ____

Age: ____

Course: ____

STRING TASKS

Task 6: Welcome Message

Print: Welcome

Task 7: String Length

Use len() to find length.

Task 8: First & Last Character

Print first and last letter.

Task 9: Upper & Lower Case

Use upper() and lower()

Task 10: Full Name

Combine first and last name.

Task 11: Repeat String

Print string 3 times.

Task 12: String Type

Use type()

STRING INDEXING TASKS

Task 13: First 3 Characters

Print using indexing.

Task 14: Last Character

Use text[-1]

Task 15: Middle Character

Use `len(text)//2`

Task 16: First & Last Characters

Print both.

Task 17: Index Practice

From "Computer" print C, m, r

Task 18: Last Two Characters

Use negative indexing

Task 19: Index Error

`text = "Hi"`

`print(text[5])`

FINAL MINI CHALLENGE

Take name as input and print:

- First letter
 - Last letter
 - Length of name
-